

QUEST-SVD: Query-Side Token Compression via Attention-Guided Sink Removal for Efficient Multimodal Late Interaction

Huy Hoang Nguyen^{1,2,*}, Duc Dung Nguyen^{1,2,*}, Duc Nam Thai^{1,2,*},
Danh Thai Hoang^{1,2}, and Thanh Huong Le^{1,2,✉}

¹ Hanoi University of Science and Technology, Hanoi, Viet Nam

² Natural Language Processing Laboratory, BK.AI Center, Hanoi, Viet Nam
{Hoang.NH226041,Dung.ND2416681,Nam.TD225053,Thai.HD235997}@sis.hust.edu.vn
huonglt@soict.hust.edu.vn

Abstract. Late-interaction retrievers for visual documents preserve fine-grained token matches, but their online scoring cost scales directly with query length. Existing efficiency methods mainly compress document representations, leaving query-side redundancy underexplored. We propose QUEST-SVD, a training-free framework that compresses query token sequences before retrieval without model fine-tuning or index modification. QUEST-SVD removes the dominant attention-sink component via Singular Value Decomposition, then preserves residual semantics through importance-ranked anchors and semantic cluster pooling. On MMDocIR and ViDoRe-V3, QUEST-SVD matches or exceeds the uncompressed baseline in several settings while retaining only 40–60% of query tokens. It reduces late-interaction FLOPs by up to 2×—a measured 1.57× end-to-end query-latency reduction at 50% retention—and gives the strongest effectiveness–efficiency trade-off among the training-free query-side and document-side baselines evaluated here. QUEST-SVD offers a practical query-side complement to existing document-side compression pipelines.

Keywords: Visual Document Retrieval · Information Retrieval · Late Interaction · Query Optimization · Singular Value Decomposition.

1 Introduction

Visual Document Retrieval (VDR) aims to retrieve relevant document pages by leveraging both textual content and visual structure. It is central to applications such as enterprise search and retrieval-augmented generation (RAG) [13], where documents often contain complex layouts, tables, and figures that text alone cannot faithfully capture [6, 16]. In real-world RAG systems, query latency has become a key bottleneck because late-interaction retrievers must score many patch tokens for each document page. OCR-based pipelines further suffer from

* These three authors contributed equally to this work.

recognition errors and loss of layout information, motivating end-to-end methods that operate directly on rendered document images.

Recent Vision-Language Models (VLMs) support this end-to-end retrieval paradigm by encoding documents as either global embeddings (single-vector) or token-level representations (multi-vector). Late-interaction models [10] are especially effective because they preserve fine-grained correspondences. Their main drawback is computational: matching cost scales with query length, making query tokens a direct source of inference latency.

Most efficiency work has focused on the document side [3, 11, 18], reducing the number of document tokens considered at query time. This strategy is query-agnostic and may discard evidence whose relevance appears only during inference. By contrast, query representations contain static redundancy, such as prompt templates, stopwords, and subword fragments, that can be compressed online without rebuilding the index. These low-importance tokens can also introduce noisy matches to irrelevant document regions.

A natural solution is to estimate query-token importance from self-attention. Yet raw attention is biased by the attention sink phenomenon [1, 22], where a small set of tokens absorbs attention mass regardless of semantic relevance. This creates a dominant low-rank pattern in attention maps, distorts importance estimation, and can lead to poor pruning decisions.

In this work, we revisit query token redundancy from this spectral perspective and propose **QUEST-SVD** (*Query-Side Token Compression via SVD Sink Removal*), a training-free framework for multimodal late-interaction retrieval. QUEST-SVD first aggregates self-attention maps and removes their dominant rank-1 component via Singular Value Decomposition (SVD), suppressing the global attention sink and producing more faithful token importance estimates. It then applies **Semantic Cluster Pooling**, which keeps high-importance tokens as anchors and consolidates low-importance tokens into them by importance-weighted pooling. This preserves residual semantics without increasing representation size or adding trainable parameters.

On MMDocIR and ViDoRe-V3, QUEST-SVD compares favorably with the training-free query-side baselines considered here, preserving baseline-level accuracy in several settings at less than two-thirds of the original query length. Our contributions are:

1. We take an early step toward query-side token optimization in multimodal late-interaction retrieval.
2. We propose a spectral importance estimator that mitigates attention sink bias via rank-1 SVD removal.
3. We introduce Semantic Cluster Pooling, an importance-weighted aggregation that preserves fine-grained semantics.
4. We demonstrate improved effectiveness–efficiency trade-offs across multiple benchmarks and retrievers.

2 Related Work

2.1 Visual Document Retrieval

Visually rich documents motivate retrieval systems beyond plain text. Traditional OCR-based lexical or dense pipelines [9, 19] flatten documents and discard layout cues. This limitation has led to *visual document retrieval* (VDR), where models operate directly on rendered pages [6, 16].

Recent VDR approaches use either single-vector or multi-vector representations. Single-vector models such as DSE [16] encode pages as global embeddings, while late-interaction models such as ColQwen2 and ColSmol [6] preserve token-level matches. Benchmarks such as ViDoRe and MMDocIR [5, 15] establish late interaction as a strong layout-aware retrieval paradigm.

2.2 Late Interaction and Efficiency

Late-interaction models, introduced by ColBERT [10], compute relevance via token-level similarity (e.g., MaxSim), improving retrieval effectiveness over single-vector methods. However, this design incurs substantial computational cost, as scoring scales with the number of query and document tokens. In visual retrieval, where a single page may produce thousands of patch tokens, this cost becomes a major bottleneck.

Most prior work addresses this challenge on the *document side*. Techniques such as residual compression and quantization in ColBERTv2 [21], PLAID indexing [20], and XTR token filtering [12] aim to reduce the number of document tokens considered at query time. Other approaches, including MUVERA [4] and token pooling methods [3], further improve efficiency by compressing or approximating document representations. All of these techniques operate on the document side, leaving query-side redundancy—which directly impacts online latency—largely unaddressed.

However, document-side compression remains structurally limited for multimodal late interaction. Patch relevance is query-dependent, so offline pruning may discard evidence whose value appears only at inference time [14]. Pooling improves storage but can dilute local cues needed by MaxSim [3, 8, 23]. Once document tokens are removed, the matching space is irreversibly constrained and changing strategy requires re-indexing [12, 17].

2.3 Token Compression and Query-Side Gap

Token pruning and compression have been widely studied in NLP and vision [2, 11]. Unlike Token Merging (ToMe) [2], which uses bipartite matching on vision transformers, QUEST-SVD selects importance-ranked anchors via purified attention and routes residual tokens by nearest-anchor cosine assignment, better preserving alignment with visual document regions. Spectral methods provide global criteria for redundancy, while [18] learns document-side token compression for ColBERT to reduce index size and accelerate retrieval.

These methods all target the document or model side, where compression is amortized offline. A key challenge for the query side is reliably estimating token importance: self-attention is biased by attention sinks [1, 22], where a few tokens dominate attention mass independent of semantic relevance.

3 Methodology

3.1 Problem Formulation

Given a query q and a corpus of document pages $\mathcal{C}$, the goal of Visual Document Retrieval (VDR) is to rank relevant pages by matching query semantics against document content and layout. Throughout this section, S_q denotes the original query length, K the compressed query length, r the retention ratio, M the number of document tokens, and d the embedding dimension.

Modern VDR systems adopt a multi-vector representation, where a Vision-Language encoder $\Phi(\cdot)$ maps inputs into sets of token embeddings:

$$\mathcal{Q} = \{\mathbf{q}_i\}_{i=1}^{S_q}, \quad \mathcal{D} = \{\mathbf{d}_j\}_{j=1}^M. \quad (1)$$

Relevance is computed via fine-grained interactions between token sets. A widely used instantiation is the late-interaction framework [6, 10], which computes:

$$S(q, \mathcal{D}) = \sum_{i=1}^{S_q} \max_j \cos(\mathbf{q}_i, \mathbf{d}_j). \quad (2)$$

While effective, this formulation incurs $O(S_q M)$ complexity, making query length a direct contributor to inference cost. In contrast to prior work that focuses on reducing M , we target the query side and seek a compressed representation $\mathcal{Q}'$, where $|\mathcal{Q}'| = K \ll S_q$, while preserving retrieval fidelity.

3.2 The QUEST-SVD Framework

We introduce **QUEST-SVD**, a training-free framework for efficient query compression in visual document retrieval. It requires no architectural change and applies to any late-interaction retriever whose encoder exposes its self-attention weights. QUEST-SVD consists of three core components: (1) robust token importance estimation via attention purification using SVD-based sink removal, (2) importance-guided semantic merging through cluster pooling and (3) weighted late-interaction scoring. The entire pipeline operates at inference time without any fine-tuning or index modification. Figure 1 gives an overview of the pipeline.

Token Importance Estimation via Attention Purification. This stage turns self-attention into a more reliable query-token saliency signal before any compression is applied. The key idea is to remove global sink-dominated attention patterns first, so that the remaining attention mass better reflects token-specific semantic contribution.

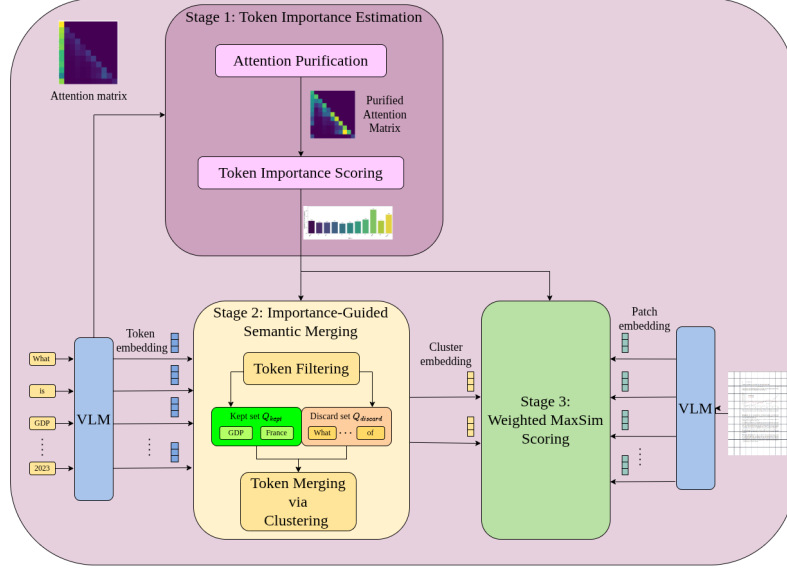


Fig. 1. Overview of the QUEST-SVD pipeline.

(1) *Attention Purification via SVD Sink Removal.* For each transformer layer l and attention head h , let $A^{(l,h)} \in \mathbb{R}^{S_q \times S_q}$ be the raw attention weight matrix, where S_q denotes the number of query tokens. We perform Singular Value Decomposition (SVD):

$$A^{(l,h)} = U \Sigma V^\top = \sum_{i=1}^{S_q} \sigma_i \mathbf{u}_i \mathbf{v}_i^\top. \quad (3)$$

The attention sink manifests as the dominant rank-1 component corresponding to the largest singular value [1]. Following recent findings [1, 22], we observe that the attention sink is highly concentrated in the leading singular component, making rank-1 removal both effective and efficient. In practice, because we only need this leading term, we estimate the first singular triplet $(\sigma_1, \mathbf{u}_1, \mathbf{v}_1)$ efficiently via two-sided power iteration [7] rather than computing the full SVD. We initialise $\mathbf{v}^{(0)}$ as the column-sum direction of $A^{(l,h)}$ — a non-negative direction that empirically retains a strong projection onto $\mathbf{v}_1$ — and alternate two matrix-vector products per iteration:

$$\mathbf{u}^{(i)} \leftarrow \frac{A^{(l,h)} \mathbf{v}^{(i-1)}}{\|A^{(l,h)} \mathbf{v}^{(i-1)}\|_2}, \quad \mathbf{v}^{(i)} \leftarrow \frac{A^{(l,h)\top} \mathbf{u}^{(i)}}{\|A^{(l,h)\top} \mathbf{u}^{(i)}\|_2}. \quad (4)$$

Convergence is guaranteed at rate $(\sigma_2/\sigma_1)^{2t}$; because attention sink matrices exhibit a large spectral gap ($\sigma_1 \gg \sigma_2$), the ratio σ_2/σ_1 is small and the residual after t iterations is negligible. Empirically the relative error in $\hat{\sigma}_1$ against an

exact SVD is already below 1% at $t = 3$ (Sect. 4.5); we use $t = 5$ throughout for margin. The rank-1 sink estimate is then:

$$A_{\text{sink}}^{(l,h)} \approx \sigma_1 \mathbf{u}_1 \mathbf{v}_1^\top. \quad (5)$$

Subtracting it and enforcing non-negativity yields a purified attention matrix that better reflects semantic interactions:

$$A_{\text{clean}}^{(l,h)} = \max(0, A^{(l,h)} - A_{\text{sink}}^{(l,h)}). \quad (6)$$

(2) *Multi-Layer Importance Aggregation.* Using the purified attention maps, we quantify each token’s importance by the amount of attention it receives from all other tokens, i.e., the column-wise sum of each purified attention matrix. For layer l and token j , the per-layer saliency is:

$$I_j^{(l)} = \frac{1}{H} \sum_{h=1}^H \sum_{i=1}^{S_q} A_{\text{clean}}^{(l,h)}(i, j), \quad (7)$$

where H is the number of attention heads.

We ℓ_∞ -normalize each layer’s importance map and apply a content mask M_{content} that excludes batch-padding positions. Query-augmentation tokens are tokenized as ordinary subwords and retained, so QUEST-SVD must suppress them through purified importance rather than explicit removal. To prioritize deeper layers (which capture higher-level semantics), we apply exponentially increasing weights:

$$\lambda_l \propto \exp\left(\frac{l-1}{L_t-1}\right), \quad l = 1, \dots, L_t, \quad (8)$$

where L_t is the number of (final) layers used for aggregation, and $\sum_l \lambda_l = 1$.

The final importance scores are obtained by weighted aggregation followed by a temperature-controlled softplus activation ($T = 0.5$):

$$I_{\text{final}} = \text{softplus}\left(\frac{1}{T} \sum_{l=1}^{L_t} \lambda_l \frac{I^{(l)}}{\max(I^{(l)})}\right) \odot M_{\text{content}}. \quad (9)$$

Softplus with $T = 0.5$ keeps importance scores smooth and strictly positive for the subsequent weighting.

Importance-Guided Semantic Merging via Cluster Pooling. Simply discarding low-importance tokens or pooling the pruned tokens may inadvertently remove valuable contextual information. Instead, we perform *importance-guided semantic merging*. After ranking tokens by I_{final} , we retain the top- K tokens as anchors; let a_k be the *original* index of the k -th anchor, so $Q_{\text{kept}} = \{\mathbf{q}_{a_k}\}_{k=1}^K$, and let Q_{disc} collect the remaining tokens, indexed by their original positions. All embeddings are ℓ_2 -normalized.

Each discarded token is assigned to its nearest anchor through hard assignment based on cosine similarity, which is equivalent to the inner product under vector normalization:

$$c(i) = \arg \max_{k \in \{1, \dots, K\}} \mathbf{q}_i^\top \mathbf{q}_{a_k}. \quad (10)$$

Let $C_k = \{i : c(i) = k\}$ be the set of discarded tokens routed to anchor k . We enrich each anchor by an importance-weighted average of itself and its assigned discarded tokens:

$$\mathbf{q}'_k = \text{normalize} \left(\frac{I_{a_k} \mathbf{q}_{a_k} + \sum_{i \in C_k} I_i \mathbf{q}_i}{I_{a_k} + \sum_{i \in C_k} I_i} \right), \quad (11)$$

where I_{a_k} and I_i are entries of I_{final} at the original token positions. The compressed query is $\mathcal{Q}' = \{\mathbf{q}'_k\}_{k=1}^K$ — exactly K embeddings, guaranteeing a strict reduction in query length while preserving residual semantic information through importance-aware aggregation.

Weighted Late-Interaction Scoring. Unlike the conventional MaxSim operator, which implicitly treats all query tokens equally, we employ a weighted MaxSim formulation to better preserve the relative importance of clustered query representations. Since both query and document embeddings are ℓ_2 -normalized, cosine similarity reduces to an inner product. The clustered query embeddings interact with the M document patch embeddings through a weighted MaxSim operator:

$$S_{\text{wt}}(\mathcal{Q}', \mathcal{D}) = \sum_{k=1}^K w_k \max_{1 \leq j \leq M} \mathbf{q}'_k{}^\top \mathbf{d}_j, \quad (12)$$

$$w_k = \frac{I_{a_k} + \sum_{i \in C_k} I_i}{\sum_{k'=1}^K (I_{a_{k'}} + \sum_{i \in C_{k'}} I_i)}.$$

Because both the clustered embeddings and the importance weights are derived from the purified multi-layer attention signal, the operator maintains high matching fidelity on a smaller query representation.

3.3 Complexity Analysis

We analyze the per-query computational cost of QUEST-SVD and compare it with the full-token baseline, focusing on the dominant cost of late interaction. Here, S_q is the query length, $K = \max(1, \lfloor rS_q \rfloor)$ is the retained query length, M is the number of document patch embeddings per page, and P is the number of candidate pages. $\mathcal{F}$ denotes computational cost in floating-point operations (FLOPs).

Baseline. Scoring P pages using the full query requires computing all pairwise dot products between query and document tokens:

$$\mathcal{F}_{\text{base}} \approx 2 S_q P M d. \quad (13)$$

This cost grows linearly with query length, corpus size, and page resolution.

QUEST-SVD. We group the online cost into the three stages of Section 3. The first two are corpus-free; only the third scales with P and M .

Stage 1: Importance estimation via attention purification. For each selected layer l and head h , QUEST-SVD processes one self-attention matrix $A^{(l,h)} \in \mathbb{R}^{S_q \times S_q}$. We extract only the leading singular triplet $(\sigma_1, \mathbf{u}_1, \mathbf{v}_1)$ via power iteration [7] (see Subsection 3.2 for algorithmic details). Each of the t iterations executes two matrix–vector products at $4S_q^2$ FLOPs; the final rank-1 reconstruction and subtraction add $2S_q^2$ FLOPs. Summing over t iterations, H heads, and L_t layers,

$$\mathcal{F}_{\text{stage1}} \approx (4t + 2) L_t H S_q^2. \quad (14)$$

This avoids a full $\mathcal{O}(S_q^3)$ SVD and keeps Stage 1 entirely in GPU-friendly matrix–vector products. The remaining operations, including clipping to obtain $A_{\text{clean}}^{(l,h)}$, computing column sums, averaging over heads, and reweighting layers, require at most $\mathcal{O}(L_t H S_q^2)$ additional work with a small constant factor. These terms are therefore omitted from the leading-order FLOP model in Eq. (17).

Stage 2: Importance-guided semantic merging. After selecting the top- K anchors, each of the $(S_q - K)$ discarded tokens is assigned to its nearest retained token. The dominant operation is the $(S_q - K) \times K$ cosine-similarity matrix, so

$$\mathcal{F}_{\text{stage2}} \approx 2(S_q - K) K d. \quad (15)$$

The subsequent weighted averaging and normalization of each cluster are lower-order in comparison.

Stage 3: Weighted late interaction. Only the final scoring stage touches the corpus. Scoring P pages using the K compressed query vectors costs

$$\mathcal{F}_{\text{stage3}} \approx 2K P M d. \quad (16)$$

Applying the scalar weights w_k does not change the leading-order cost.

The total is therefore:

$$\mathcal{F}_{\text{ours}} \approx \underbrace{(4t + 2) L_t H S_q^2}_{\text{Stage 1}} + \underbrace{2(S_q - K) K d}_{\text{Stage 2}} + \underbrace{2K P M d}_{\text{Stage 3}} \quad (17)$$

As $PM \rightarrow \infty$, the stage-3 term dominates, and the theoretical FLOP reduction approaches $1/r$. This corresponds to a $2\times$ reduction at 50% token retention and $\approx 3.3\times$ at 30% token retention. The corpus-free overhead of Stages 1 and 2 is fully amortised beyond a break-even corpus size (P^*). For our measured ColQwen2 configuration ($S_q = 41.4 \pm 9.3$ over 1,658 queries, $d = 128$, $H = 12$, $L_t = 28$, average $M \approx 613$ over 20,395 pages, $t = 5$, and 50% token retention), Eq. (17) gives $P^* \approx 4$ pages when attention is reused from an eager forward pass. Observed speedups remain below the FLOP reduction because attention materialisation and memory traffic, rather than arithmetic alone, dominate the overhead.

Since P^* is two orders of magnitude below corpus-scale retrieval ($P \geq 10^3$ pages), the query-side arithmetic overhead is negligible in every regime we evaluate, and the realised FLOP reduction is within a few percent of the $1/r$ asymptote throughout.

4 Experimental Setup

4.1 Datasets

We evaluate QUEST-SVD on two widely used benchmarks for multimodal document retrieval, covering both page-level and layout-aware settings to ensure comprehensive evaluation across different granularity levels.

MMDocIR [5] is a benchmark supporting both page-level and fine-grained layout-level retrieval. Its evaluation set contains 313 long documents paired with 1,658 expert-annotated queries across ten domains, including academic papers, financial reports, and legal documents. Its high visual complexity—18.8% images and 16.7% tables—makes it well suited for evaluating fine-grained token-level interactions.

ViDoRe-V3 [15] is a large-scale benchmark designed for multimodal retrieval in real-world RAG scenarios. It comprises 26,000 document pages and 3,099 human-verified queries, spanning diverse professional domains (e.g., finance, telecom, pharmaceuticals). The dataset provides high-quality annotations for both retrieval relevance and visual grounding, and serves as a testbed for generalisation beyond the MMDocIR development distribution; we evaluate its English and French corpora.

4.2 Evaluation Metrics

We evaluate QUEST-SVD on both retrieval effectiveness and computational efficiency. For effectiveness, we report Recall@K on MMDocIR (page and layout retrieval) and nDCG@10 on ViDoRe-V3, following the standard evaluation protocols of each benchmark. For efficiency, we measure the reduction in late-interaction cost using Floating Point Operations (FLOPs) as a hardware-agnostic proxy, which scales linearly with the number of query tokens.

4.3 Implementation Details

We evaluate two public checkpoints, ColQwen2-v1.0 and ColSmol-500M [6], which share the same 127k-pair ColPali training set and LoRA configuration ($r = \alpha = 32$) but differ in backbone family and scale. Queries are encoded with `colpali-engine` < 0.3.13, which prepends the "Query: " prefix these checkpoints were trained with. Purification uses SVD rank $k = 1$. We aggregate the final transformer layers with temperature $T = 0.5$. All methods use eager attention on an NVIDIA RTX PRO 6000 GPU to expose attention maps under a matched setting; we therefore report FLOPs as the primary hardware-agnostic efficiency metric. We evaluate random pruning and hierarchical pooling on both the query and document sides, and include document-side 1D and 2D pooling baselines for spatial token aggregation. Hierarchical Ward used a Ward distance matrix, whereas Spherical KMeans was iterated until convergence. Random pruning and other stochastic components were averaged across independent runs under matched experimental settings.

4.4 Results

We report results at 50% token retention as the primary operating point, corresponding to a $2\times$ reduction in MaxSim FLOPs. This operating point lies near the knee of the ColQwen2 MMDocIR compression–quality curve in Figure 2, beyond which all methods maintain near-baseline quality. Table 1 reports MMDocIR page / layout Recall@5 across ten domains under this setting, which measures retrieval coverage within a small candidate set.

Table 1. MMDocIR page and layout Recall@5 with ColQwen2 at 50% token retention ($2\times$ lower MaxSim FLOPs); micro avg. is over all queries. **Qry./Doc.:** query-/document-side. **RP** Random Pruning, **AP** Attention Pruning, **HP** Hierarchical Pooling, **SP** Spherical Pooling, **1D/2D-P** 1D/2D Pooling. Domains: **ACP** Academic paper, **A/I** Administration/Industry, **BR** Brochure, **FR** Financial report, **GOV** Government, **GB** Guidebook, **LAW** Laws, **NW** News, **RR** Research report, **TW** Tutorial/Workshop.

Side	Method	MMDocIR domain										Micro avg.
		ACP	A/I	BR	FR	GOV	GB	LAW	NW	RR	TW	
ColQwen2 — page Recall@5												
Baseline	Full	90.49	92.70	77.41	69.37	89.63	82.66	95.45	76.64	85.18	82.86	83.14
Qry.	RP [11]	92.30	90.92	78.17	69.88	88.73	82.37	94.69	75.18	84.22	81.22	83.17
	AP [11]	92.82	91.87	76.53	71.77	88.73	82.23	94.69	75.91	84.97	84.02	83.95
	HP [3, 8]	92.56	91.51	76.53	72.14	88.73	85.33	94.69	76.64	84.21	85.03	84.20
	SP [3, 8]	91.36	92.70	77.52	69.74	89.63	81.53	95.45	73.72	85.51	82.22	83.10
	QUEST-SVD	93.40	91.81	79.16	72.65	88.73	84.43	95.45	75.91	85.87	84.18	84.75
Doc.	RP [11]	89.07	83.18	73.68	68.02	87.61	80.57	93.18	75.18	84.66	82.47	81.76
	1D-P [24]	90.63	87.56	78.40	67.32	89.41	82.62	94.70	75.18	83.52	80.63	83.00
	2D-P [24]	89.92	92.92	74.12	66.23	89.41	82.67	93.94	74.45	83.95	82.31	82.99
	HP [3, 8]	90.88	92.70	74.12	69.19	88.51	82.26	96.21	73.72	85.03	82.95	83.56
ColQwen2 — layout Recall@5												
Baseline	Full	63.52	35.53	41.95	49.20	73.44	57.88	69.89	63.15	58.43	58.56	58.46
Qry.	RP [11]	58.91	33.56	37.46	48.40	67.99	59.25	68.31	58.03	54.87	60.11	55.81
	AP [11]	60.11	34.62	39.01	45.17	70.73	55.55	67.51	59.49	55.63	55.34	55.29
	HP [3, 8]	62.15	34.29	43.44	50.60	70.04	56.83	67.34	60.95	55.97	54.22	57.21
	SP [3, 8]	61.45	30.42	39.89	48.39	74.17	53.45	69.93	58.76	56.67	57.04	56.62
	QUEST-SVD	65.81	33.00	37.70	50.00	74.42	45.19	70.08	59.13	59.32	60.59	58.24
Doc.	RP [11]	57.23	32.43	41.62	40.26	72.75	53.48	69.30	58.76	52.80	53.92	51.30
	1D-P [24]	59.15	30.50	41.21	46.67	70.90	54.27	65.79	57.42	54.34	59.43	53.06
	2D-P [24]	51.59	24.26	40.65	35.05	60.77	43.31	63.49	50.00	44.75	50.10	46.70
	HP [3, 8]	57.45	32.78	36.99	40.83	72.00	46.57	63.55	21.97	45.68	38.32	47.43

On ColQwen2, QUEST-SVD achieves the highest average layout Recall@5 of 58.24 among all compressed methods, coming within 0.22 points of the uncompressed baseline (58.46) at 50% token retention ($2\times$ lower MaxSim FLOPs). It edges out the next-best query-side method, Hierarchical Pooling (57.21), by 1.03 points, and outperforms the best document-side method, 1D-Pooling (53.06), by a 5.18-point margin. Moreover, QUEST-SVD improves the average page Recall@5 to 84.75, surpassing the full baseline (83.14), and exceeds the uncompressed baseline on six out of ten domains.

The same superior trend holds on ColSmol-500M, where QUEST-SVD obtains the highest average layout score among query-side approaches at 47.87, narrowly ahead of Hierarchical Pooling (47.27) and the best document-side method

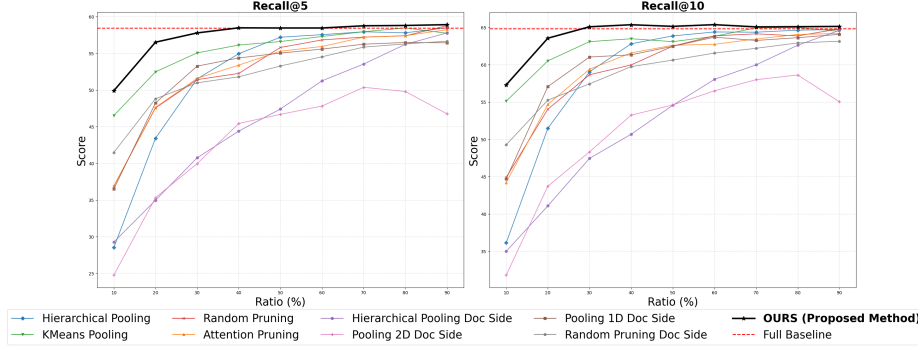


Fig. 2. Recall@5 and Recall@10 vs. token-retention ratio on MMDocIR layout retrieval with ColQwen2. The dashed line denotes the uncompressed baseline. Curves are distinguished by marker and line style as well as colour.

Hierarchical Pooling (45.59); the per-domain pattern mirrors ColQwen2, with the largest gains on table-intensive documents. As single-domain results carry higher variance, we treat these as indicative rather than conclusive.

Figure 2 extends this comparison across the full retention range on MM-DocIR layout retrieval with ColQwen2. QUEST-SVD exceeds the uncompressed baseline at 30% token retention on R@10 ($\approx 3.33\times$ lower FLOPs) and 40% token retention on R@5 ($\approx 2.5\times$); no competing method reaches baseline parity below 70% token retention, and neither Random nor Attention Pruning recovers to baseline at any tested ratio. The advantage is most pronounced under extreme compression ($r \leq 20\%$), where QUEST-SVD remains 2–5 points ahead of the best competing pooling methods across our experiments, suggesting that SVD-ranked selection preserves semantically critical token dimensions that heuristic pruning and merging criteria fail to protect.

To assess whether these gains generalise beyond the development distribution, we evaluate on ViDoRe-V3 [15], which spans ten corpora in English and French, none of which overlap with the MMDocIR evaluation set. Both checkpoints are trained on the same English-only ColPali set, which is verified to share no document with ViDoRe (verified for ViDoRe V1; both checkpoints predate V3), so ViDoRe-V3 is held out for both; its French corpora are additionally zero-shot. The two curves therefore vary backbone family and scale while holding training data fixed. As shown in Figure 3, QUEST-SVD is the only evaluated method that achieves performance comparable to the uncompressed baseline across both backbones under a fixed-index setting, matching the 45.23 nDCG@10 ColQwen2 baseline at both 50% and 60% token retention. A plausible mechanism is that purified attention keeps query terms tied to entities, operations, and information needs, which are more transferable across corpora than surface-level prompt or filler tokens.

A qualitative inspection suggests why. After sink removal, high-scoring tokens are typically named entities, action-bearing verbs, and words expressing

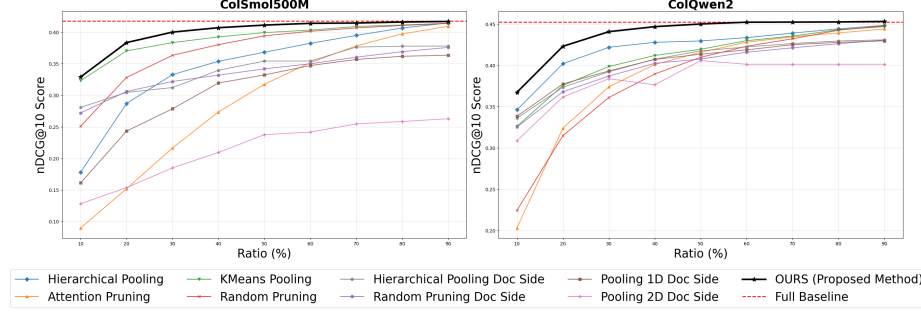


Fig. 3. nDCG@10 vs. token-retention ratio on ViDoRe-V3. Curves are distinguished by marker and line style as well as colour.

retrieval intent, whereas prompt-template fragments, stopwords, and weak function tokens score low. This selectivity preserves the query terms that trigger meaningful page-level matches. Three trends hold across both benchmarks and both backbones: query-side compression is more favorable than document-side compression in our late-interaction setting; QUEST-SVD gives the best observed effectiveness–efficiency trade-off among the query-side methods evaluated; and in several domains it turns compression into a small net quality gain at lower MaxSim FLOPs.

4.5 Ablation Studies

We conduct ablation studies on MMDocIR Layout with ColQwen2 at a fixed 50% token-retention setting, an operating point that enforces substantial compression ($\approx 2\times$ fewer MaxSim FLOPs) while remaining well within the regime where compressed query representations retain discriminative structure. We combine five design factors in a controlled grid: (i) the number of final transformer layers used for importance aggregation ($n \in \{4, 8, 16, 28\}$), (ii) importance-weighted scoring, (iii) SVD-based sink purification, (iv) an explicit first-token-removal (FTR) baseline serving as a hard-pruning proxy for spectral purification, and (v) cluster-based pooling.

We also measure the online latency of each ablated configuration over a retrieval workload of 164,838 layout embeddings, so the reported times capture both the fixed query-side compression overhead and the downstream MaxSim scoring cost at realistic corpus scale.

Table 2 reports retrieval quality and latency for the main QUEST-SVD design choices at 50% token retention. The full configuration, combining SVD-based purification, cluster pooling, and importance-weighted scoring at $n=28$, achieves the strongest compressed R@1, R@10, and nDCG@10 scores (36.48, 65.23, and 47.51, respectively), while its R@5 of 58.24 remains close to the best compressed result of 58.35 at $n=16$.

The joint latency measurements show that these effectiveness gains preserve the computational advantage of query compression. At $n=28$, the full configuration reduces end-to-end latency from 55.58 ms to 35.35 ms, corresponding to a $1.57\times$ speedup at 50% token retention. Recall@10 reaches 65.23, slightly above the full-token reference of 64.84, while nDCG@10 (47.51) remains within 0.5 points of the uncompressed baseline (47.98). This operating point therefore provides a substantially more favorable efficiency–effectiveness trade-off.

Table 2. Ablation results on MMDocIR Layout with ColQwen2 at a fixed 50% token-retention ratio. Experiments are conducted on an NVIDIA RTX PRO 6000 GPU, and latency reports the average per-query time. In the “Scoring” column, **T** denotes traditional MaxSim and **W** denotes importance-weighted MaxSim. The Ratio column indicates speed relative to the no-compression baseline (values >1 are faster).

n	Design Choices				Retrieval Metrics				Latency	
	Scoring	SVD	FTR	Cluster pool.	R@1	R@5	R@10	nDCG@10	Time (ms)	Ratio
4	T	✓	×	✓	35.27	56.73	63.65	45.98	29.22	$1.90\times$
4	W	✓	×	✓	35.98	58.09	64.41	46.89	29.22	$1.90\times$
8	T	✓	×	✓	35.26	56.72	64.22	46.56	30.25	$1.84\times$
8	W	✓	×	✓	36.26	58.16	64.72	46.98	30.25	$1.84\times$
16	T	✓	×	✓	36.26	57.30	64.94	47.04	32.30	$1.72\times$
16	W	✓	×	✓	36.38	58.35	65.03	47.38	32.30	$1.72\times$
28	T	✓	×	✓	36.24	58.06	65.09	47.14	35.35	$1.57\times$
28	W	✓	×	✓	36.48	58.24	65.23	47.51	35.35	$1.57\times$
28	W	×	×	✓	35.04	57.48	63.96	46.04	29.31	$1.90\times$
28	W	×	✓	✓	35.50	57.42	64.63	46.47	29.74	$1.87\times$
28	W	✓	×	×	36.03	57.63	64.69	46.89	35.34	$1.57\times$
28	W	×	×	×	27.62	48.80	56.46	38.73	29.24	$1.90\times$
<i>Baseline (no compression)</i>					36.29	58.46	64.84	47.98	55.58	$1.00\times$

Removing SVD purification reduces nDCG@10 from 47.51 to 46.04, while removing cluster pooling reduces it to 46.89. Removing both components yields the largest degradation, with nDCG@10 dropping to 38.73.

The first-token-removal baseline (nDCG@10 = 46.47) partially closes this gap but remains inferior to full SVD purification, suggesting that SVD-based purification provides a more effective mechanism for mitigating attention sink effects. Weighted scoring also provides consistent gains over traditional MaxSim across all aggregation depths, suggesting that the compression-derived importance weights carry useful ranking information beyond token selection alone.

Spectral Analysis of the Attention Sink. Figure 4 tests the low-rank structure that the purification step of Sect. 3.2 assumes. The leading singular component dominates the spectrum outside the first two layers: the rank-1 energy fraction rises steeply over the first three layers and then stays above 65%, averaging 78.86% across all 28 layers, so the final n layers that QUEST-SVD aggregates all sit in this high-energy regime.

The spectral gap is large on average ($\sigma_1/\sigma_2 = 23.75$; median across layers ≈ 13.5) but very unevenly distributed: two layers exceed 100 while most stay

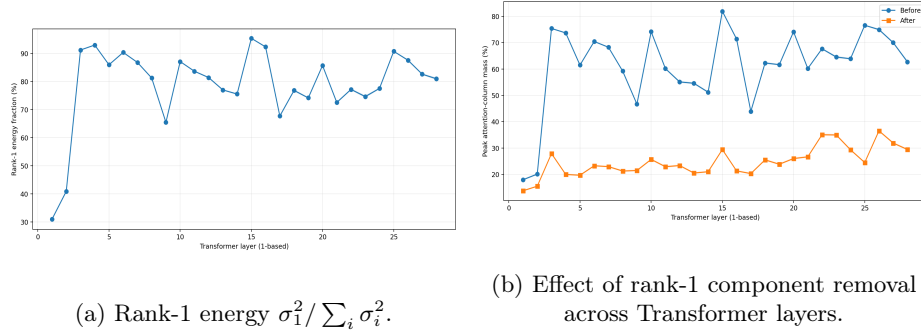


Fig. 4. Spectral diagnostics of content-only query self-attention in ColQwen2, over 1,658 MMDocIR queries, all 28 layers and all 12 heads. Markers are per-layer means over queries and heads; shaded bands are 95% confidence intervals and are usually narrower than the marker. Note the differing vertical scales.

below 25, so the mean overstates the gap at a typical layer and it is the smallest-gap layers that bound convergence. That bound is still comfortable: the relative error in $\hat{\sigma}_1$ against an exact SVD falls from 1.57% at $t = 1$ to 0.17% at $t = 3$ and 0.054% at $t = 5$, and at $t = 5$ it is below 1% for 98.34% of layer-head instances. These diagnostics show that content-only query attention is strongly low-rank and that our rank-1 estimate is numerically reliable. Direct evidence further supports the sink interpretation: the rank-1 component concentrates 73.5% of its own column mass on a single position, that position is the first sequence position in 95.9% of layer-head instances, and after removal the residual’s peak column share falls to 24.8%. Consistent with this, Table 2 shows that removing SVD purification costs 1.47 nDCG@10 points while the first-token-removal proxy recovers less than a third of that gap — the sink is a globally organised low-rank mode, not a single-token artefact.

5 Conclusion

We presented QUEST-SVD, a training-free query-side compression framework for multimodal late-interaction retrieval. By purifying self-attention maps via rank-1 SVD sink removal and consolidating residual tokens into importance-weighted cluster anchors, QUEST-SVD halves MaxSim FLOPs at 50% token retention—a measured $1.57\times$ end-to-end query-latency reduction—while exceeding the uncompressed baseline on six of ten MMDocIR domains (ColQwen2, page Recall@5) and matching it on ViDoRe-V3 across both backbones. These results show that late-interaction query representations contain substantial exploitable redundancy that can be compressed online while leaving the underlying retriever and document collection unchanged.

Two limitations bound these results. First, QUEST-SVD reads per-layer, per-head attention weights, so it requires an attention implementation that materialises them; fused kernels such as FlashAttention do not expose these maps,

and our reported latencies use eager attention on both the compressed and uncompressed sides. Second, we have evaluated primarily on multimodal visual document retrieval. Future work will extend the evaluation to a wider range of tasks and domains, including text-only retrieval on benchmarks such as MS MARCO and BEIR, and will investigate importance estimators that avoid explicit attention materialisation.

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